

EX03:Hand Washing Detection using Accelerometer Signals

The objective of this exercise is that you analyse the code provided and make the link with the curse. You have to provide a short report that comments and analyse the results. You can use directly the results or adapt them to you needs.

import the numerical library

```
In [1]: import numpy as np
# import signal processing library
import scipy.signal as sp
# import plotting library
import pylab as py
py.ion()
py.close('all')
```

load the ecg signal

```
In [2]: x = np.genfromtxt('accel.dat')
# sampling frequency of the signal is 500 Hz
fs = 40
# generate correponding time vector
t = np.arange(len(x))/fs
```

The signal is an acceleration signal measured at the wrist.

The signal records the acceleration during the hand washing protocol in an hospital. The goal is to detect the hand washing sequence.

The hand washing protocol produced rhythmical movements.

The protocol is known to take place between 20 and 30 seconds.

Plot time signal.

Q: Comment the figure.

The acceleration signal remains relatively steady during the first ~20 seconds, showing only small fluctuations around a constant baseline. At around 20 seconds a marked transition occurs: the amplitude increases sharply and the waveform becomes clearly rhythmic, indicating the start of the hand-washing sequence. Between ~22 and ~30 seconds the activity is sustained, with frequent bursts and pronounced excursions, such as the large drop near 30 seconds, that reflect the vigorous and repetitive strokes typical of the washing protocol. After 30 seconds the signal amplitude decreases and its variability reduces, marking the conclusion of the hand-washing protocol. This temporal

pattern clearly isolates the washing sequence as a distinct, high-energy episode embedded within two other quieter signal segments.

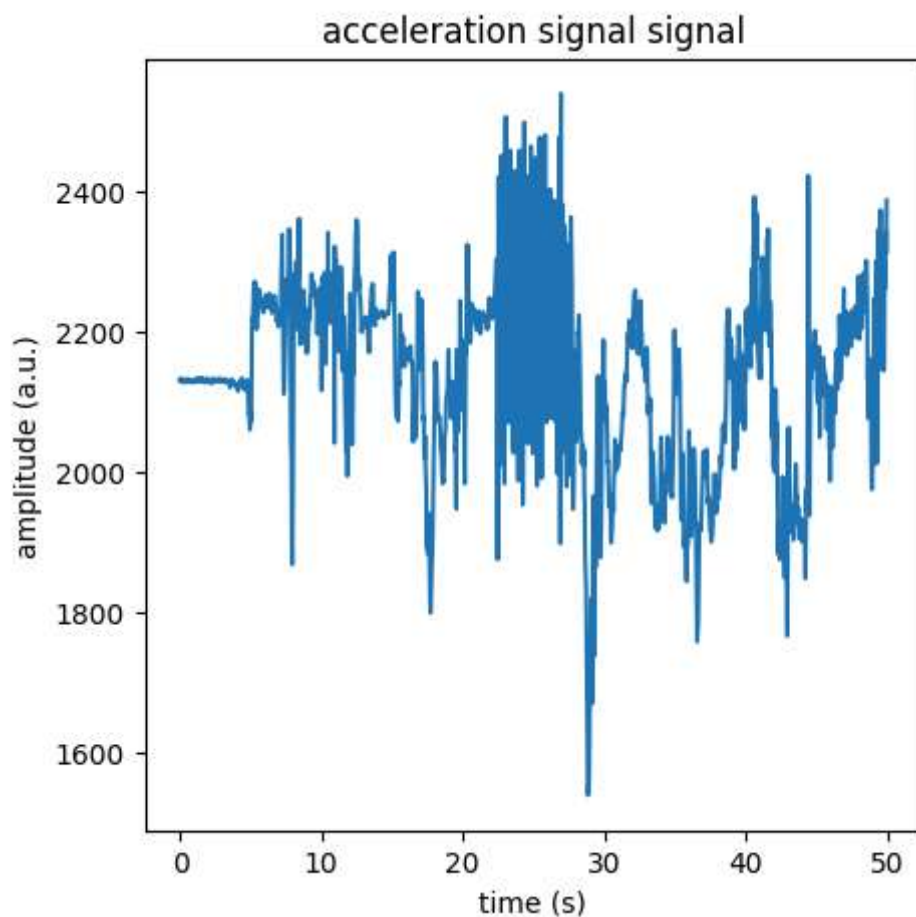
Compute the FFT of the signal

```
In [3]: x_fft = np.fft.fft(x)
# Determine the frequency scale
f_fft = np.arange(len(x_fft))/len(x_fft)*fs
```

plot the signal

```
In [4]: py.figure(1, figsize=[5,5])
py.clf()
py.plot(t, x)
py.xlabel('time (s)')
py.ylabel('amplitude (a.u.)')
py.title('acceleration signal signal')
```

```
Out[4]: Text(0.5, 1.0, 'acceleration signal signal')
```



High pass the signal.

Q: Comment the figure.

After applying a high-pass filter at 0.5 Hz, the wrist acceleration signal changes noticeably. The raw signal contained both fast hand movements and slow variations due to posture, sensor noise, or gravity, which made the hand washing sequence harder to see. The filter removes these slow components and highlights the faster, rhythmic ones. In the top plot, between 20 and 30 seconds, the signal shows a clearer oscillatory pattern,

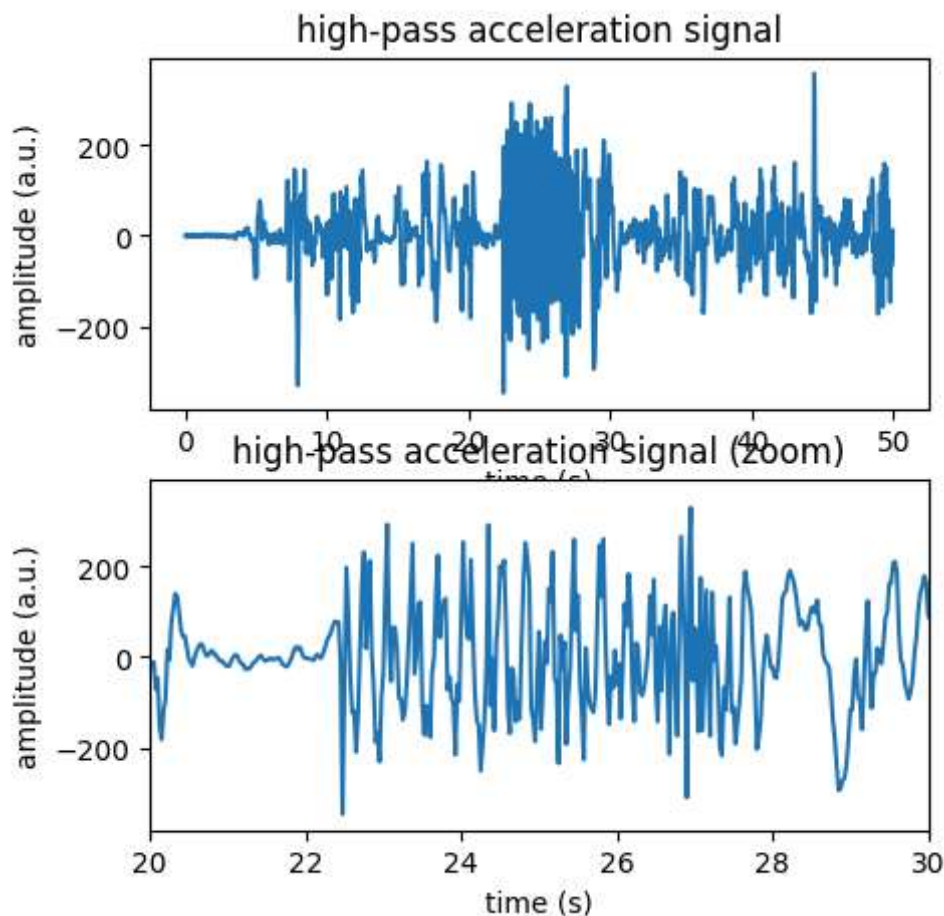
corresponding to hand washing. The zoomed view further emphasizes this periodicity with compact, regular oscillations. In short, the high-pass filter cleans the signal from low-frequency noise and brings out the hand washing rhythm, making it much easier to detect.

high-pass filter with cutoff frequency of 0.5 Hz

```
In [5]: b, a = sp.butter(4, 0.5/fs*2, btype='high')
# zero-phase filtering of the signal
x_hp = sp.filtfilt(b, a, x)
```

```
In [6]: py.figure(2, figsize=[5,5])
py.clf()
py.subplot(2,1,1)
py.plot(t, x_hp)
py.xlabel('time (s)')
py.ylabel('amplitude (a.u.)')
py.title('high-pass acceleration signal')
py.subplot(2,1,2)
py.plot(t, x_hp)
py.xlabel('time (s)')
py.ylabel('amplitude (a.u.)')
py.title('high-pass acceleration signal (zoom)')
py.xlim(20, 30)
```

Out[6]: (20.0, 30.0)



Band pass the signal between 2.4 and 3.2 Hz.

Q: Based on previous figure, comment the selection of the frequencies.

The band-pass filtered signal between 2.4 and 3.2 Hz represents the dominant oscillatory component of the wrist accelerometer during the handwashing task. This frequency range corresponds to the repetitive wrist and hand movements that occur during scrubbing, rinsing, and rubbing, typically performed at a rate of about 2–3 cycles per second. After filtering, only the components of the signal within this frequency band are preserved, meaning that slow movements (e.g., posture changes, drift, or low-frequency noise) and high-frequency components (e.g., tremors, sensor noise, or sharp micro-motions) are suppressed. What remains is the periodic rhythmic motion of the wrist, which characterizes the core mechanical pattern of handwashing.

Q: Why zero phase filter (filtfilt) is used?

A zero-phase filter such as `filtfilt` is applied to avoid phase distortion. When a causal filter is used in the forward direction only, the output is delayed by the filter's group delay, causing a temporal shift of peaks and relevant features. By filtering the signal forward and backward, this delay is cancelled and the overall phase response becomes zero, ensuring that the morphology and timing of the hand-washing signal are preserved.

Analogic limit of the passband frequency

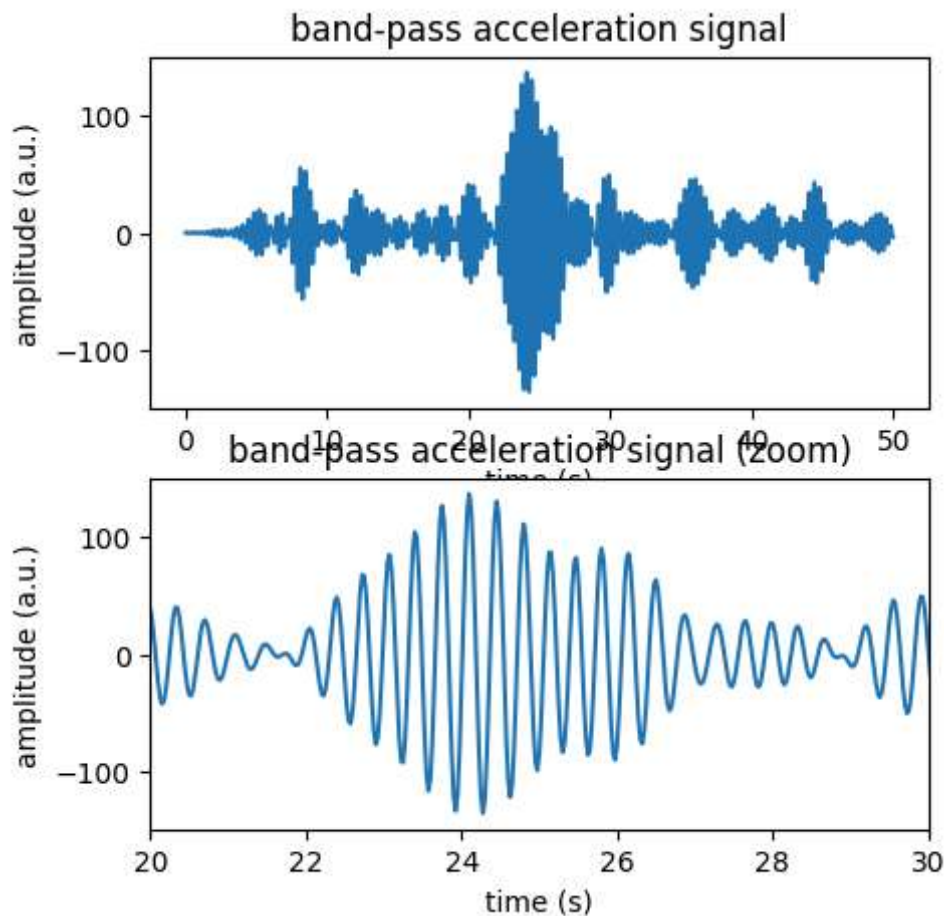
```
In [7]: f_pass = np.array([2.4, 3.2])
# Analogic limit of the stopband frequency
f_stop = np.array([0, 5])
# Conversion into Nyquist frequency
f_pass_N = f_pass/fs*2
f_stop_N = f_stop/fs*2
# Max attenuation in passband (dB)
g_pass = 3
# Min attenuation in stopband (dB)
g_stop = 40
# Determine the order and the cutoff frequency of a butterworth filter
ord, wn = sp.buttord(f_pass_N, f_stop_N, g_pass, g_stop)
# Compute the coefficients of the filter
b, a = sp.butter(ord, wn, btype='band')
# Filter the signal
x_bp = sp.filtfilt(b, a, x_hp)
```

```
c:\Users\alexb\AppData\Local\Programs\Python\Python313\Lib\site-packages\scipy\signal\_filter_design.py:4137: RuntimeWarning: divide by zero encountered in divide
    nat = ((stopb ** 2 - passb[0] * passb[1]) /
```

```
In [8]: py.figure(3, figsize=[5,5])
py.clf()
py.subplot(2,1,1)
py.plot(t, x_bp)
py.xlabel('time (s)')
py.ylabel('amplitude (a.u.)')
py.title('band-pass acceleration signal')
py.subplot(2,1,2)
py.plot(t, x_bp)
py.xlabel('time (s)')
py.ylabel('amplitude (a.u.)')
```

```
py.title('band-pass acceleration signal (zoom)')
py.xlim(20, 30)
```

Out[8]: (20.0, 30.0)



Low-pass filter of the power of the band-pass signal.

Q: Why use the power of the acceleration signal?

When the wrist acceleration signal is observed directly, it appears as a waveform oscillating around zero with both positive and negative values. This makes it difficult to assess the intensity of the motor activity, since positive and negative contributions tend to cancel each other out. By computing the power of the signal (i.e., by squaring it), an always positive measure is obtained that directly reflects the energy of the movement. This representation makes it easier to identify periods of greater activity, such as the hand washing sequence, which corresponds to a clear increase in power compared to intervals of rest or minor movements.

Q: How the detection of hand washing is obtained?

The detection process begins with the band-pass filtered acceleration signal, in order to isolate the frequencies typical of the rhythmic hand washing movement. The power of this signal is then computed, providing an instantaneous measure of movement intensity. Since the raw power remains noisy, a low-pass filter is applied to smooth the trend and produce a more regular curve. In this filtered curve, hand washing appears as a clear and localized peak. At this stage, a threshold is defined: when the power exceeds the threshold, the system detects that hand washing is taking place; when the power falls

below the threshold, the activity is considered finished. In the plot, this detection is represented by the orange line, which rises clearly between 20 and 30 seconds, exactly during the interval of hand washing

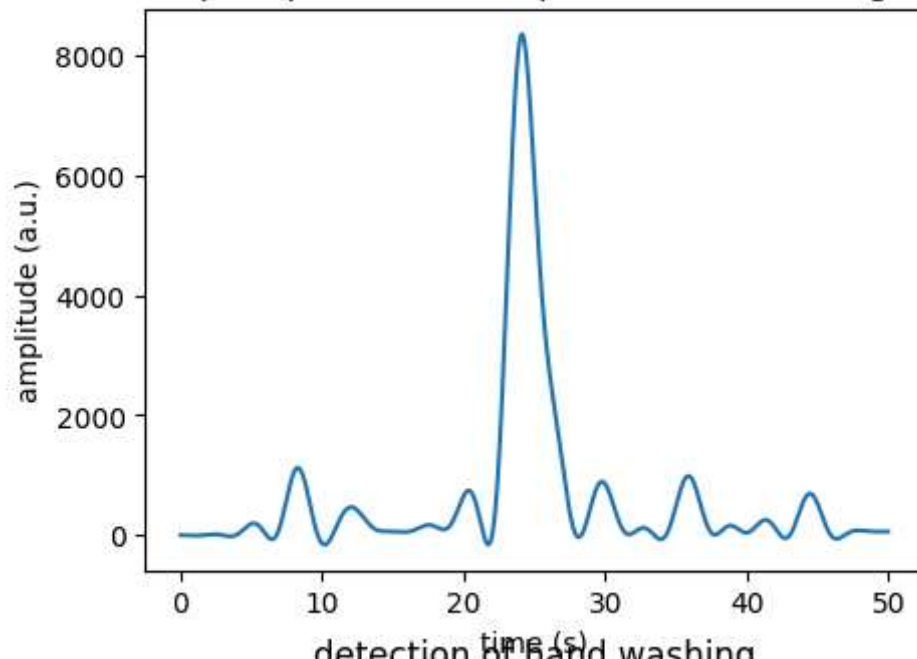
Analogic limit of the passband frequency

```
In [9]: f_pass = 0.4
# Analogic limit of the stopband frequency
f_stop = 0.8
# Conversion into Nyquist frequency
f_pass_N = f_pass/fs*2
f_stop_N = f_stop/fs*2
# Max attenuation in passband (dB)
g_pass = 3
# Min attenuation in stopband (dB)
g_stop = 40
# Determine the order and the cutoff frequency of a butterworth filter
ord, wn = sp.buttord(f_pass_N, f_stop_N, g_pass, g_stop)
# Compute the coefficients of the filter
b, a = sp.butter(ord, wn)
# Filter the signal
x_pow = sp.filtfilt(b, a, x_bp**2)
# detection
det = x_pow > 2000
```

```
In [10]: py.figure(4, figsize=[5,8])
py.clf()
py.subplot(2,1,1)
py.plot(t, x_pow)
py.xlabel('time (s)')
py.ylabel('amplitude (a.u.)')
py.title('low-pass power of band-pass acceleration signal')
py.subplot(2,1,2)
py.plot(t, x_hp, label='acceleration')
py.plot(t, det*200, linewidth=2, label='detection')
py.xlabel('time (s)')
py.ylabel('amplitude (a.u.)')
py.title('detection of hand washing')
py.legend(loc='upper right')
```

```
Out[10]: <matplotlib.legend.Legend at 0x220b0ea4980>
```

low-pass power of band-pass acceleration signal



detection of hand washing

